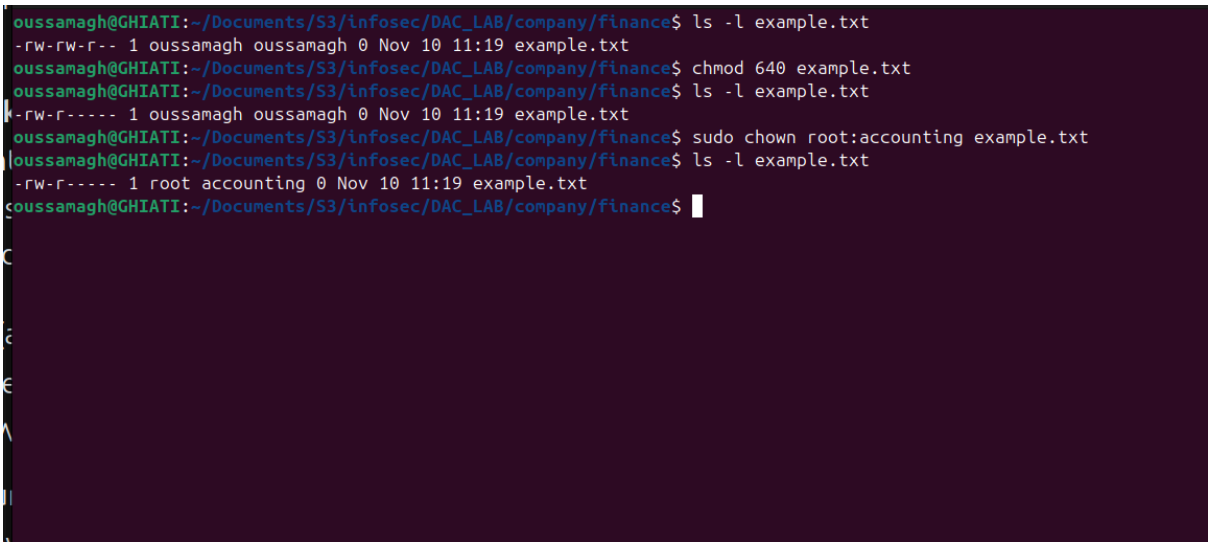



Question 6: No, while the directory (750) has read + execute permissions to the group, the file (600) restricts the read permissions to the owner only.

## 3. MAC

### SELinux:




1.

| | Document_A (Secret) | Document_B (Confidential) | Document_C (Public) |
|---|---|---|---|
| Alice (Secret) | yes | yes | yes |
| Bob (Confidential) | No | yes | yes |
| Charlie (Public) | No | No | yes |

2.

Yes Bob can read Document B.

3.

No Alice can not write to Document_C (No_write down)

4.

The information leakage is prevented by no_write_down and no_read_up, so a used that can read secret information can never write it to public or confidential clearance, also the users with low clearance won t be able to read what s in the secret level (no_read_up).

5.

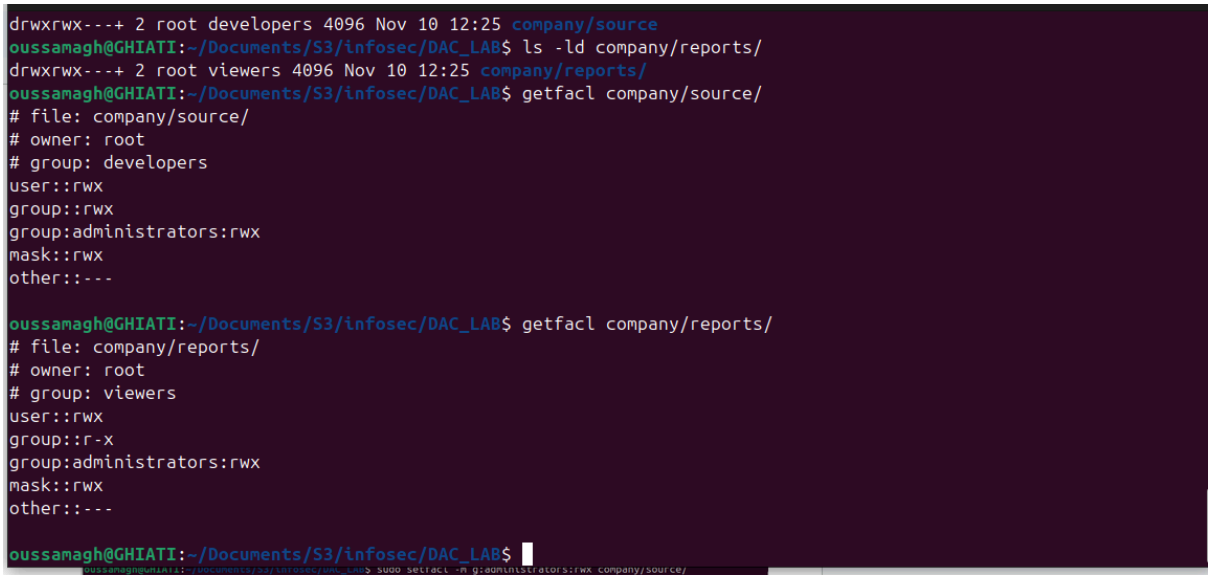
No, under MAC the clearance is only a necessary condition, so the owner's explicit DAC denial still blocks access, since both DAC and MAC must allow a request before it s granted.

## 4 . RBAC:

```
oussamagh@GHIATI:~/Documents/S3/infosec/DAC_LAB/company/finance$ sudo usermod -aG developers alice
oussamagh@GHIATI:~/Documents/S3/infosec/DAC_LAB/company/finance$ sudo usermod -aG viewers bob
oussamagh@GHIATI:~/Documents/S3/infosec/DAC_LAB/company/finance$ sudo usermod -aG administrators charlie
oussamagh@GHIATI:~/Documents/S3/infosec/DAC_LAB/company/finance$ groups  alice
alice : alice users developers
oussamagh@GHIATI:~/Documents/S3/infosec/DAC_LAB/company/finance$ groups bob
bob : bob users viewers
oussamagh@GHIATI:~/Documents/S3/infosec/DAC_LAB/company/finance$ groups charlie
charlie : charlie users administrators
oussamagh@GHIATI:~/Documents/S3/infosec/DAC_LAB/company/finance$ cd ..
oussamagh@GHIATI:~/Documents/S3/infosec/DAC_LAB/company$ cd ..
oussamagh@GHIATI:~/Documents/S3/infosec/DAC_LAB$ sudo mkdir -p company/source
oussamagh@GHIATI:~/Documents/S3/infosec/DAC_LAB$ sudo mkdir -p company/reports
oussamagh@GHIATI:~/Documents/S3/infosec/DAC_LAB$ sudo chown root:developers company/source/
oussamagh@GHIATI:~/Documents/S3/infosec/DAC_LAB$ sudo chmod 770 company/source/
oussamagh@GHIATI:~/Documents/S3/infosec/DAC_LAB$ sudo chown root:viewers company/reports/
oussamagh@GHIATI:~/Documents/S3/infosec/DAC_LAB$ sudo chmod 750 company/reports
oussamagh@GHIATI:~/Documents/S3/infosec/DAC_LAB$
```

1.

```
oussamagh@GHIATI:~/Documents/S3/infosec/DAC_LAB/company/finance$ sudo usermod -aG developers alice
oussamagh@GHIATI:~/Documents/S3/infosec/DAC_LAB/company/finance$ sudo usermod -aG viewers bob
oussamagh@GHIATI:~/Documents/S3/infosec/DAC_LAB/company/finance$ sudo usermod -aG administrators charlie
oussamagh@GHIATI:~/Documents/S3/infosec/DAC_LAB/company/finance$ groups  alice
alice : alice users developers
oussamagh@GHIATI:~/Documents/S3/infosec/DAC_LAB/company/finance$ groups bob
bob : bob users viewers
oussamagh@GHIATI:~/Documents/S3/infosec/DAC_LAB/company/finance$ groups charlie
charlie : charlie users administrators
oussamagh@GHIATI:~/Documents/S3/infosec/DAC_LAB/company/finance$ cd ..
oussamagh@GHIATI:~/Documents/S3/infosec/DAC_LAB/company$ cd ..
oussamagh@GHIATI:~/Documents/S3/infosec/DAC_LAB$ sudo mkdir -p company/source
oussamagh@GHIATI:~/Documents/S3/infosec/DAC_LAB$ sudo mkdir -p company/reports
oussamagh@GHIATI:~/Documents/S3/infosec/DAC_LAB$ sudo chown root:developers company/source/
oussamagh@GHIATI:~/Documents/S3/infosec/DAC_LAB$ sudo chmod 770 company/source/
oussamagh@GHIATI:~/Documents/S3/infosec/DAC_LAB$ sudo chown root:viewers company/reports/
oussamagh@GHIATI:~/Documents/S3/infosec/DAC_LAB$ sudo chmod 750 company/reports
oussamagh@GHIATI:~/Documents/S3/infosec/DAC_LAB$ sudo setfacl -m g:administrators:rwx company/source/
oussamagh@GHIATI:~/Documents/S3/infosec/DAC_LAB$ sudo setfacl -m g:administrators:rwx company/reports
oussamagh@GHIATI:~/Documents/S3/infosec/DAC_LAB$ ls -ld company/source
drwxrwx---+ 2 root developers 4096 Nov 10 12:25 company/source
oussamagh@GHIATI:~/Documents/S3/infosec/DAC_LAB$ ls -ld company/reports/
drwxrwx---+ 2 root viewers 4096 Nov 10 12:25 company/reports/
oussamagh@GHIATI:~/Documents/S3/infosec/DAC_LAB$ getfacl company/source/
```




1,2,3,4 : All the required tasks have been successfully completed. The three groups **(administrators, developers, and viewers)** were created, and the users **Alice, Bob, and Charlie** were assigned their respective roles. Permissions and ownership were configured for both **the source code and reports directories**. The initial configuration was analyzed, and the missing **administrator permissions** were identified and corrected using ACLs. Finally, the access policy was implemented to ensure that each role has the appropriate level of access according to the company's requirements.

## 5. ABAC:

1. Decisions - Alice: Granted (department matches, clearance sufficient, within 08:00-18:00). - Bob: Denied. - Charlie: Denied. - David: Denied.

2. Attribute responsible for each denial - Bob: department (HR does not match Finance). - Charlie: clearance (Public is lower than Secret). - David: time (22:00 is outside 08:00-18:00).

3. Alice requests an HR document Denied. Her department (Finance) does not match the document's department (HR), even though her clearance and the time are fine.

4. David at 17:00 Yes, the decision changes to granted. His department matches, his Secret clearance is sufficient, and 17:00 is within working hours.

5. Why ABAC is more flexible than RBAC ABAC evaluates attributes of the user, the document, and the context (such as time) at request time, so a single policy covers every combination and the decision can change with the

situation, as David's did. RBAC only knows static roles, so expressing "same department + sufficient clearance + working hours" would need a separate role for every combination (role explosion), and it cannot natively check time or other context.